ELECTRICAL / AUTOMATION / SYSTEMS / CASE STUDY

INFRA RESILIENCE

CASE STUDY

03

A benchmark workflow for separating the initiating fault from the alarm storm it causes.

PROPOSED RESEARCH / NOT YET VERIFIED

THE BUSINESS CONTEXT

A material handling site runs conveyors with drives and occupancy sensors and controller logic. When one fault occurs dozens of alarms follow within seconds. Engineers sift logs by hand to find the origin and the same incident can be explained differently by different people.

WHO IS INVOLVED AND WHAT THEY NEED

01	**Controls engineer**	Needs a ranked list of likely causes with the evidence behind each.
02	**Maintenance lead**	Wants to send a technician to the right place first time.
03	**Research reviewer**	Needs a fair test of diagnosis against threshold alarms alone.

SCOPE AND HONEST LIMITS

- ⚠ A simulated plant cannot validate certified industrial protection.
- ⚠ Correlation is not by itself a causal explanation.
- ⚠ No accuracy has been measured yet and all targets are for testing.
- ⚠ Real sites have faults and wiring that the model will not capture.

CURRENT WORKFLOW AND FAILURE POINTS

03

PROPOSED RESEARCH / NOT YET VERIFIED

HOW THE WORK RUNS TODAY

1 TODAY

Alarm flood

One fault raises many alarms across drives and sensors and software layers.

2 TODAY

Manual log search

An engineer reads timestamps across several systems to guess the start.

3 TODAY

Threshold alarms only

Alarms report symptoms and say little about which came first.

4 TODAY

Informal diagnosis

The explanation depends on whoever happens to be on shift.

5 TODAY

Fix and move on

The incident is closed with a short note and no replayable trace.

6 TODAY

No scoring

Nobody records whether the first diagnosis was right.

WHERE IT BREAKS

Cause hidden in symptoms	Downstream alarms drown the single fault that started the cascade.
Inconsistent diagnosis	Two engineers can name different root causes for one incident.
Missing telemetry	Gaps in data force guesses that are never flagged as guesses.
No learning loop	Past incidents are not kept in a form that can test new methods.

REDESIGNED WORKFLOW AND DECISION AUTHORITY

03

PROPOSED RESEARCH / NOT YET VERIFIED

THE REDESIGNED WORKFLOW

1 PERSON

Define plant model

Specify drives and sensors and logic and alarms and the dependencies between them.

→

2 SYSTEM

Inject known faults

Simulate single and simultaneous faults and record the true initiating cause.

→

3 SYSTEM

Capture telemetry

Log timestamped events with chosen gaps removed to mimic missing data.

↓

4 SYSTEM

Rank candidate causes

Score causes against evidence and show the supporting alarms for each.

→

5 PERSON

Engineer review

An engineer inspects the ranking and the ambiguity and confirms or overrides.

→

6 SYSTEM

Score and replay

Compare with injected truth and store the trace for later replay.

SYSTEM STEP PERSON DECISION

WHERE AUTHORITY STAYS WITH A PERSON

Acceptance of a root cause	CONTROLS ENGINEER	A ranking is advice and the engineer owns the call to act on it.
Any change to plant or protection	MAINTENANCE LEAD	Physical intervention needs a named person and normal site procedures.
Benchmark scenarios and scoring rules	RESEARCH REVIEWER	Independent rules stop the method being tuned to flatter itself.

ARCHITECTURE, DATA AND CONTROLS

03

PROPOSED RESEARCH / NOT YET VERIFIED

SYSTEM ARCHITECTURE

1	**Process model**	Discrete event simulation of conveyors and drives and sensors with controller logic.
2	**Telemetry and logs**	Timestamped state transitions and alarms stored as replayable traces.
3	**Fault injector**	Scripted single and multiple faults across motors and sensors and timing.
4	**Causal ranking**	Dependency graph or probabilistic scoring that links symptoms to causes.
5	**Benchmark harness**	Records injected truth and observations and the ranked explanation for scoring.

CORE DATA ENTITIES

- **Plant component**
- **Fault scenario**
- **Event trace**
- **Alarm**
- **Candidate cause**
- **Diagnosis score**

RISKS AND THE CONTROL FOR EACH

Correlation read as cause →	Rankings must show the dependency path and supporting events and not co occurrence alone.
Overconfident ranking →	Ambiguous cases are reported as ties with their uncertainty and not forced to one answer.
Method tuned to scenarios →	Scenarios are split into development and held out sets before any scoring.
Simulation mistaken for site proof →	Reports state that no live protection or procedure is validated by the benchmark.

EVALUATION AND DELIVERY PLAN

03

PROPOSED RESEARCH / NOT YET VERIFIED

HOW SUCCESS IS MEASURED

Top 1 accuracy	Share of scenarios where the injected initiating fault is ranked first.
Top 3 accuracy	Share of scenarios where the injected fault appears in the first three.
Detection latency	Time from fault onset to the first useful ranked explanation.
False alarm rate	Spurious fault reports per period of healthy simulated running.
Degradation under gaps	Change in accuracy as telemetry is removed or faults overlap.

TEST PLAN

1. **Replay a stored trace and confirm the same ranking is produced each time.**
2. **Check plant invariants such as no item occupying two sensors at once.**
3. **Compare threshold only diagnosis with ranked diagnosis on identical scenarios.**
4. **Remove telemetry in steps and confirm uncertainty rises rather than hiding.**

DELIVERY PLAN

01 ALIGN	02 UNDERSTAND	03 DESIGN	04 DELIVER	05 LAUNCH
Agree the system to model and the faults that matter and how success is scored.	Map dependencies between components with those who run similar systems.	Specify the simulator and event format and the ranking approach and benchmark rules.	Build the plant and injector and ranker and run the benchmark with reviewers.	Publish the dataset and replay library and the evaluation report with limits.
DECISION Scope of plant and faults	**DECISION** Which dependencies are real	**DECISION** Scoring rules sign off	**DECISION** Accept or reject findings	**DECISION** What is released

DELIVERABLES

- **Plant and controller simulation specification**
- **Fault scenario benchmark dataset**
- **Invariant tests for plant and controller logic**
- **Replay library of event traces**
- **Root cause evaluation report**